

Lucas Narloch Zabla

Software Engineer | Backend & Cloud Systems

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Computer Science student and Backend Developer with nearly two years of experience building and operating production software on AWS. Experienced with Python, distributed and event-driven systems, SQL and NoSQL databases, automated testing, CI/CD, infrastructure as code, observability, and production troubleshooting. Hands-on computer science foundations through compiler and systems programming projects in C.

TECHNICAL SKILLS

- **Languages:** Python, C, TypeScript, SQL
- **CS Fundamentals:** Data Structures & Algorithms, Complexity Analysis, Object-Oriented Programming, Operating Systems, Computer Networks
- **Backend & Distributed Systems:** REST APIs, Event-Driven Architecture, Amazon SQS, AWS Lambda, MQTT
- **Databases:** PostgreSQL, MySQL, DynamoDB
- **Cloud & DevOps:** AWS, Docker, Terraform, GitHub Actions, CI/CD, Linux, Bash, Git
- **Testing & Reliability:** pytest, Testcontainers, Unit Testing, Integration Testing, Observability, Production Troubleshooting

EXPERIENCE

Backend Developer - SmartLy Fabricação de Dispositivos Inteligentes Ltda.

10/2026 - Present

- Lead the consolidation of multiple AWS Lambda functions into a single FastAPI API organized into services and repositories, reducing deployment time from approximately 12 to 5 minutes and latency on several endpoints from 7–10 seconds to approximately 1.5 seconds (79–85%).
- Redesigned AWS network architecture, replacing multiple VPC endpoints with a NAT Gateway and adding an EC2 bastion for database access, reducing monthly network costs by approximately 64% (US\$90 to US\$32) and expanding Lambda connectivity.
- I am restructuring the MySQL schema of a production IoT platform, remodeling entities, relationships, and migration flows to better reflect the domain and preserve integrity and compatibility during transition.

Backend Developer Intern - SmartLy Fabricação de Dispositivos Inteligentes Ltda.

10/2024 - 09/2026

- Built an IoT data migration endpoint across MySQL, DynamoDB, S3, and IoT Core via MQTT, supporting thousands of historical records per operation.
- Built 120 integration tests and 300 unit tests with pytest and Testcontainers, reducing production regression risk.
- Analyzed Lambda error metrics in CloudWatch and fixed unhandled edge cases, reducing average errors from 113 to 5 per five-minute interval.
- Introduced asynchronous, event-driven processing with Amazon SQS, handling peaks above 40,000 notifications per day and removing blocking operations from the main flow.

PROJECTS

PocScript

Programming language and compiler implemented in C, covering lexical analysis, AST parsing, semantic analysis, custom intermediate representation, LLVM IR emission, and native executable generation. Includes a Linux syscall-based runtime, static type system, scope resolution, arrays, pointers, and 196 unit and integration tests.

Technologies: C, LLVM IR, NASM, Make, Linux system calls, Compiler Design

[Code](#)

Sunlog.dev

Production web platform with 33 REST endpoints, PostgreSQL, JWT authentication, asynchronous processing, and more than 100 automated tests. Deployed on AWS using Docker and CI/CD pipelines, with structured logging and monitoring through Prometheus and Grafana.

Technologies: NestJS, TypeScript, PostgreSQL, Docker, AWS, GitHub Actions, Prometheus, Grafana

[Code](#) | [Live demo](#)

EDUCATION

Bachelor of Computer Science - PUCPR

02/2025 - 12/2028

Full scholarship recipient.

Relevant Coursework: Algorithms, OOP, operating systems, networks, databases, software engineering.

LANGUAGES

- **Portuguese:** Native
- **English:** Professional Working Proficiency